1. a) The first two columns are dependent and therefore the matrix is not invertible. **[2]**

b) $\det(A) = 1 - a$ and therefore, in order for A to be invertible then $a \neq 1$. **[2]**

c) The required volume is $\det\left(\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 1 & 1 & 5 \end{bmatrix}\right) = 4$ **[2]**

d) $\det(A^T) = \det(A) = \det(-A) = (-1)^n \det(A)$. If n is an odd positive integer then $(-1)^n = -1$ and therefore, $\det(A) = -\det(A) \Rightarrow \det(A) = 0$. This means that the matrix is not invertible. **[4]**

e) $\det(5A) = 5^3 \det(A) = 1250$

$\det(3A^{-1}) = \dfrac{3^3}{\det(A)} = \dfrac{27}{10}$

$\det(3A^3) = 3^3[\det(A)]^3 = 27 \times 1000 = 2700$

$\det[2(A^T)^{-1}] = 2^3 \det[(A^T)^{-1}] = 2^3 \dfrac{1}{\det(A^T)} = \dfrac{2^3}{\det(A)} = \dfrac{8}{10}$

$\det(B) = -\det(A) = -10$ since B is obtained by transposing A and making a single row swap. **[4]**

f) We select the first orthogonal direction to be $A = a = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$ and $\|A\| = \sqrt{6}$. Therefore,

$q_1 = \dfrac{A}{\|A\|} = \begin{bmatrix} 1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}$. The second direction is:

$$B = b - \frac{AA^T}{A^T A} b = \begin{bmatrix} -2 \\ 2 \\ -2 \end{bmatrix} - \frac{1}{6}\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}\begin{bmatrix} 1 & -1 & 2 \end{bmatrix}\begin{bmatrix} -2 \\ 2 \\ -2 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \\ -2 \end{bmatrix} - \frac{1}{6}\begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ 2 & -2 & 4 \end{bmatrix}\begin{bmatrix} -2 \\ 2 \\ -2 \end{bmatrix} =$$

$$\begin{bmatrix} -2 \\ 2 \\ -2 \end{bmatrix} - \frac{1}{6}\begin{bmatrix} -8 \\ 8 \\ -16 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} -4 \\ 4 \\ 4 \end{bmatrix}$$

$\|B\| = \dfrac{1}{6}\sqrt{3 \times 16} = \dfrac{4\sqrt{3}}{6}$

$$q_2 = \frac{B}{\|B\|} = \frac{6}{4\sqrt{3}}\frac{1}{6}\begin{bmatrix} -4 \\ 4 \\ 4 \end{bmatrix} = \frac{1}{\sqrt{3}}\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$\frac{BB^T}{B^T B} = \frac{1}{3}\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}\begin{bmatrix} -1 & 1 & 1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

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$$\frac{BB^T}{B^TB}c=\frac{1}{3}\begin{bmatrix}1&-1&-1\\-1&1&1\\-1&1&1\end{bmatrix}\begin{bmatrix}-2\\3\\-2\end{bmatrix}=\frac{1}{3}\begin{bmatrix}-3\\3\\3\end{bmatrix}$$

$$\frac{AA^T}{A^TA}=\frac{1}{6}\begin{bmatrix}1\\-1\\2\end{bmatrix}\begin{bmatrix}1&-1&2\end{bmatrix}=\frac{1}{6}\begin{bmatrix}1&-1&2\\-1&1&-2\\2&-2&4\end{bmatrix}$$

$$\frac{AA^T}{A^TA}c=\frac{1}{6}\begin{bmatrix}1&-1&2\\-1&1&-2\\2&-2&4\end{bmatrix}\begin{bmatrix}-2\\3\\-2\end{bmatrix}=\frac{1}{6}\begin{bmatrix}-9\\9\\-18\end{bmatrix}=\frac{1}{2}\begin{bmatrix}-3\\3\\-6\end{bmatrix}$$

$$\frac{BB^T}{B^TB}c+\frac{AA^T}{A^TA}c=\frac{1}{2}\begin{bmatrix}-5\\5\\-4\end{bmatrix}$$

$$c-\frac{BB^T}{B^TB}c-\frac{AA^T}{A^TA}c=\begin{bmatrix}-2\\3\\-2\end{bmatrix}-\frac{1}{2}\begin{bmatrix}-5\\5\\-4\end{bmatrix}=\frac{1}{2}\begin{bmatrix}1\\1\\0\end{bmatrix}$$

$$\|c\|=\frac{1}{2}\sqrt{2}$$

$$q_3=\frac{c}{\|c\|}=\frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\\0\end{bmatrix}$$

Therefore, $Q=\begin{bmatrix}\frac{1}{\sqrt{6}}&\frac{-1}{\sqrt{3}}&\frac{1}{\sqrt{2}}\\-\frac{1}{\sqrt{6}}&\frac{1}{\sqrt{3}}&\frac{1}{\sqrt{2}}\\\frac{2}{\sqrt{6}}&\frac{1}{\sqrt{3}}&\frac{1}{\sqrt{2}}\end{bmatrix}$

$$q_1^Ta=\sqrt{6},\ q_1^Tb=-\frac{8}{\sqrt{6}},\ q_1^Tc=-\frac{9}{\sqrt{6}},\ q_2^Tb=\frac{2}{\sqrt{3}},\ q_2^Tc=\frac{3}{\sqrt{3}},\ q_3^Tc=\frac{1}{\sqrt{2}}$$

$$R=\begin{bmatrix}\sqrt{6}&\frac{-8}{\sqrt{6}}&\frac{-9}{\sqrt{6}}\\0&\frac{2}{\sqrt{3}}&\frac{3}{\sqrt{3}}\\0&0&\frac{1}{\sqrt{2}}\end{bmatrix}$$

The solution of the system $Fx=b$ is given now through the system $Rx=Q^Tb$

[6]

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2. a)

(i) The sequential steps of elimination give the following intermediate matrices:

$$A=\begin{bmatrix}1&1&-1&2&3&3\\2&1&-2&0&3&4\\1&0&-1&-2&0&1\\3&2&-3&2&6&7\end{bmatrix} \text{ and then } \begin{bmatrix}1&1&-1&2&3&3\\2&1&-2&0&3&4\\1&0&-1&-2&0&1\\0&-1&0&-4&-3&-2\end{bmatrix},$$

$$\begin{bmatrix}1&1&-1&2&3&3\\2&1&-2&0&3&4\\0&-1&0&-4&-3&-2\\0&-1&0&-4&-3&-2\end{bmatrix}, \begin{bmatrix}1&1&-1&2&3&3\\2&1&-2&0&3&4\\0&-1&0&-4&-3&-2\\0&0&0&0&0&0\end{bmatrix},$$

$$\begin{bmatrix}1&1&-1&2&3&3\\0&-1&0&-4&-3&-2\\0&-1&0&-4&-3&-2\\0&0&0&0&0&0\end{bmatrix} \text{ and finally } R=\begin{bmatrix}1&1&-1&2&3&3\\0&-1&0&-4&-3&-2\\0&0&0&0&0&0\\0&0&0&0&0&0\end{bmatrix}$$

We know that $EA=R$ with

$$E=\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&-1&1&0\\0&0&0&1\end{bmatrix}\cdot\begin{bmatrix}1&0&0&0\\-2&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix}\cdot\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&-1&1\end{bmatrix}\cdot\begin{bmatrix}1&0&0&0\\0&1&0&0\\-1&0&1&0\\0&0&0&1\end{bmatrix}\cdot$$

$$\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\-3&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0\\-2&1&0&0\\1&-1&1&0\\-2&0&-1&1\end{bmatrix}.$$

The row space has dimension 2. A basis of the row space can be formed by the first two rows of R. **[2]**

(ii) The nullspace has dimension 6-2=4. We choose the free variables to be the ones which correspond to columns 3 to 6. We then find the four special solutions. These consist a basis of the nullspace. They are: $[1\ 0\ 1\ 0\ 0\ 0]^T$, $[2\ -4\ 0\ 1\ 0\ 0]^T$, $[0\ -3\ 0\ 0\ 1\ 0]^T$, $[-1\ -2\ 0\ 0\ 0\ 1]^T$ **[2]**

(iii) The column space has the same dimension as the row space, i.e., 2. We can choose any two independent columns for example the first two, to form a basis for the column space. **[2]**

(iv) $EA=R$ and the last two rows of R are zero and therefore, the last 2 rows of E form a basis of the left nullspace which has dimension is 2. **[2]**

b)

(i) FALSE since the maximum number of independent vectors in R^n is n. **[1]**

(ii) TRUE the zero vector is dependent to all vectors. **[1]**

(iii) TRUE since if two vectors in T were dependent then two vectors in S would be dependent. **[1]**

(iv) TRUE obvious **[1]**

(v) FALSE if the rows are more than the columns and columns are independent the system might not have a solution. **[2]**

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c)

(i) We have a set of equations

$$C-2D=0$$
$$C-D=0$$
$$C=1$$
$$C+D=1$$

and therefore the system is

$$\begin{bmatrix}1 & -2\\ 1 & -1\\ 1 & 0\\ 1 & 1\end{bmatrix}\begin{bmatrix}C\\ D\end{bmatrix}=\begin{bmatrix}0\\ 0\\ 1\\ 1\end{bmatrix}$$

The system doesn't have a solution since the solutions that is obtained from 2 of the equations doesn't satisfy the rest. **[2]**

(ii) Instead of solving the system $\begin{bmatrix}1 & -2\\ 1 & -1\\ 1 & 0\\ 1 & 1\end{bmatrix}\begin{bmatrix}C\\ D\end{bmatrix}=\begin{bmatrix}0\\ 0\\ 1\\ 1\end{bmatrix}$ we solve the system

$$\begin{bmatrix}1 & 1 & 1 & 1\\ -2 & -1 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & -2\\ 1 & -1\\ 1 & 0\\ 1 & 1\end{bmatrix}\begin{bmatrix}C\\ D\end{bmatrix}=\begin{bmatrix}1 & 1 & 1 & 1\\ -2 & -1 & 0 & 1\end{bmatrix}\begin{bmatrix}0\\ 0\\ 1\\ 1\end{bmatrix}\Rightarrow\begin{bmatrix}4 & -2\\ -2 & 6\end{bmatrix}\begin{bmatrix}C\\ D\end{bmatrix}=\begin{bmatrix}2\\ 1\end{bmatrix}\Rightarrow$$

$$\begin{bmatrix}C\\ D\end{bmatrix}=\frac{1}{20}\begin{bmatrix}6 & 2\\ 2 & 4\end{bmatrix}\begin{bmatrix}2\\ 1\end{bmatrix}\Rightarrow C=\frac{7}{10}, D=\frac{4}{5}$$

The straight line is $7/10+4t/5$. **[2]**

(iii) The error vector is

$$e=\begin{bmatrix}0\\ 0\\ 1\\ 1\end{bmatrix}-\begin{bmatrix}1 & -2\\ 1 & -1\\ 1 & 0\\ 1 & 1\end{bmatrix}\begin{bmatrix}\frac{7}{10}\\ \frac{4}{5}\end{bmatrix}=\begin{bmatrix}\frac{9}{10}\\ \frac{1}{10}\\ \frac{3}{10}\\ \frac{1}{2}\end{bmatrix}$$

[2]

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3. a) For $\lambda = 0$ the characteristic polynomial is 7. Therefore, 0 is not a root of the characteristic polynomial in which case the matrix is invertible. **[2]**

b) In this case $S = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$ and $\Lambda = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$. $S^{-1} = -\begin{bmatrix} 1 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$

Therefore, $A = S\Lambda S^{-1} = \begin{bmatrix} -7 & 10 \\ -5 & 8 \end{bmatrix}$, $A^3 = S\Lambda^3 S^{-1} = \begin{bmatrix} -43 & 70 \\ -35 & 62 \end{bmatrix}$,

$$A^3 \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} -43 & 70 \\ -35 & 62 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 38 \\ 46 \end{bmatrix}$$ **[2]**

c) Consider the matrix A: The eigenvalues are 4,4,2,2 and therefore the matrix has repeated eigenvalues. It might or might not be diagonalizable. The eigenvectors are $\begin{bmatrix} 2 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$ which are independent. Therefore, the matrix is diagonalizable. **[6]**

d)

(i) For matrix A we have

$$\begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \Rightarrow \begin{matrix} x+3y+3z=x \\ -3x-5y-3z=y \\ 3x+3y+z=z \end{matrix} \Rightarrow \begin{matrix} 3y+3z=0 \\ -3x-6y-3z=0 \\ 3x+3y=0 \end{matrix} \Rightarrow$$

$$\begin{matrix} y+z=0 \\ x+y=0 \end{matrix} \Rightarrow \begin{matrix} z=-y \\ x=-y=z \end{matrix}$$

An eigenvector can be $\begin{bmatrix} 1 & -1 & 1 \end{bmatrix}^T$

$$\begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -2 \begin{bmatrix} x \\ y \\ z \end{bmatrix} \Rightarrow \begin{matrix} x+3y+3z=-2x \\ -3x-5y-3z=-2y \\ 3x+3y+z=-2z \end{matrix} \Rightarrow \begin{matrix} 3x+3y+3z=0 \\ -3x-3y-3z=0 \\ 3x+3y+3z=0 \end{matrix} \Rightarrow$$

$x+y+z=0 \Rightarrow z=-x-y$

The eigenvectors are $\begin{bmatrix} x & y & -x-y \end{bmatrix}^T = x\begin{bmatrix} 1 & 0 & -1 \end{bmatrix}^T + y\begin{bmatrix} 0 & 1 & -1 \end{bmatrix}^T$ **[3]**

(ii) For matrix B we have

$$\begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \Rightarrow \begin{matrix} 2x+4y+3z=x \\ -4x-6y-3z=y \\ 3x+3y+z=z \end{matrix} \Rightarrow \begin{matrix} x+4y+3z=0 \\ -4x-7y-3z=0 \\ 3x+3y=0 \end{matrix} \Rightarrow$$

$$\begin{matrix} y+z=0 \\ x+y=0 \end{matrix} \Rightarrow \begin{matrix} z=-y \\ x=-y=z \end{matrix}$$

An eigenvector can be $\begin{bmatrix} 1 & -1 & 1 \end{bmatrix}^T$

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$$\begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = -2\begin{bmatrix} x \\ y \\ z \end{bmatrix} \Rightarrow \begin{matrix} 2x+4y+3z=-2x \\ -4x-6y-3z=-2y \\ 3x+3y+z=-2z \end{matrix} \Rightarrow \begin{matrix} 4x+4y+3z=0 \\ -4x-4y-3z=0 \\ 3x+3y+3z=0 \end{matrix} \Rightarrow$$

$$\begin{matrix} x+y+z=0 \\ 4x+4y+3z=0 \end{matrix} \Rightarrow \begin{matrix} x+y=0 \\ z=0 \end{matrix}$$

An eigenvector can be $\begin{bmatrix} 1 & -1 & 0 \end{bmatrix}^T$ **[3]**

(iii) A is diagonalizable because it has 3 independent eigenvectors but B is not. **[2]**

(iv) $\Lambda = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$ $S = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & -1 & -1 \end{bmatrix}$ $S^{-1} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & -1 \\ 1 & 2 & 1 \end{bmatrix}$ and $A = S\Lambda S^{-1}$ **[2]**

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